IDEAS - Institute of Data Engineering, Analytics and Science Foundation

Summer Internship Program 2026

Project Title: Exploratory Data Analysis with Sales Data

Project Notebook

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Question 1: Import Data and Store-wise Revenue Analysis (4 Marks)

Import `pandas` and `numpy`. Download the dataset `sales.csv` from https://drive.google.com/drive/folders/1gieHICVDBbUKMZiSF4YRQMAJic-w50JM?usp=sharing. Load the dataset `sales.csv` into a DataFrame. Then, find the top 5 stores with the highest total sales and the bottom 5 stores with the lowest total sales.

Tasks:

- Aggregate total sales by store_id
- Rank stores by total revenue
- Compare average sales per department within those stores

In [ ]:
```
# Write your answer here
```

Question 2: Department Performance Consistency (3 Marks)

Which departments have the most stable weekly sales and which are the most volatile?

Tasks:

- Use standard deviation and coefficient of variation
- Identify highly stable and highly volatile departments

In [ ]:
```
# Write your answer here
```

Question 3: Holiday vs Non-Holiday Sales (3 Marks)

Analyze whether holidays significantly affect weekly sales.

Tasks:

- Compare average sales during holidays vs non-holidays
- Calculate percentage increase/decrease
- Identify departments benefiting most from holidays

In [ ]:
```
# Write your answer here
```

Question 4: Seasonal Trend Detection (3 Marks)

Identify whether sales increase or decrease during specific months.

Tasks:

- Extract month from the date column
- Compute average monthly sales
- Determine highest and lowest sales months

In [ ]:
```
# Write your answer here
```

Question 5: Store and Department Contribution (4 Marks)

Which combinations of store and department contribute the most to overall revenue?

Tasks:

- Group using store_id and department
- Find top and worst-performing combinations
- Compute percentage contribution

In [ ]:
```
# Write your answer here
```

Question 6: Load MNIST Data (3 Marks)

Load the MNIST dataset using `sklearn.datasets.load_digits`. Separate the dataset into features (`X`) and target labels (`y`). Print the shape of the features and the target arrays.

Expected Output: The shape of `X` and `y`.

In [ ]:
```
# Write your answer here
```

Question 7: K-Means Clustering (5 Marks)

Import `KMeans` from `sklearn.cluster`. Initialize and fit the K-Means clustering model on the MNIST features (`X`). Since we know there are 10 digits (0-9), set the number of clusters to 10. Assign the cluster labels to a variable `kmeans_labels`.

Expected Output: A successfully fitted K-Means model and the cluster labels array.

In [ ]:
```
# Write your answer here
```

Question 8: F1 Score Evaluation for Clustering (5 Marks)

Evaluate the clustering performance using the F1 score. Since K-Means assigns arbitrary cluster labels, you will first need to map each cluster label to the most frequent true label in that cluster. After mapping the labels, calculate and print the macro-averaged F1 score using `sklearn.metrics.f1_score`.

Expected Output: The calculated F1 score of the clustering.

In [ ]:
```
# Write your answer here
```